AL-NEELAIN UNIVERSITY

Faculty of Computer Science and Information Technology — Graduate Studies

COSC726: AGENTIC ARTIFICIAL INTELLIGENCE (2026)

FINAL COURSEWORK PROJECT DELIVERABLE (30% MODULE WEIGHT)

AGENTIC AI CLINICAL DECISION SUPPORT SYSTEM FOR DIABETES

A Multi-Agent Framework with Calibrated Probabilistic Modeling, ADA 2024/2026 Clinical Reasoning, and Human-in-the-Loop Quality Gate Governance

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Domain Category	Health (Clinical Decision Support Systems — CDSS)
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Deliverable 1 (GitHub URL)	https://github.com/MohamedAbulgasim/COSC726-Agentic-AI-Diabetes-CDSS
Deliverable 2 (YouTube URL)	https://youtu.be/W_mOF3TpEwc (Unlisted, 1080p, 10:00 Max)
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1. Problem and Domain Fit

1.1 The Clinical Problem in Diabetes Care

Diabetes mellitus represents one of the most critical global public health emergencies of the 21st century. According to the International Diabetes Federation (IDF) and the World Health Organization (WHO), over 537 million adults currently live with diabetes, a figure projected to surge past 783 million by 2045. Almost half of all individuals living with type 2 diabetes remain undiagnosed until microvascular (retinopathy, nephropathy, neuropathy) or macrovascular (coronary artery disease, stroke) complications have already inflicted irreversible tissue damage. Early, reliable, and accessible risk stratification is therefore paramount to preserving quality of life and curbing healthcare expenditure.

Historically, machine learning research has addressed diabetes prediction using benchmark datasets such as the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) Pima Indians Diabetes database. However, existing academic literature and clinical practice reveal severe limitations in conventional approaches. Most existing applications treat diabetes prediction as an isolated binary classification task run through monolithic, black-box algorithms (e.g. standard Random Forests or Support Vector Machines). While these systems report high theoretical accuracy on curated test splits, they fail drastically when exposed to realistic clinical environments. They suffer from uncalibrated risk probabilities, biological impossibility tolerance (accepting zero values for blood pressure or glucose without remediation), zero explanatory transparency, and an absence of clinical safety governance.

1.2 Academic Pedigree: Evolution from Previous Semester Machine Learning Project

To properly evaluate the architectural leap presented in this submission, it is essential to trace its academic origin within the postgraduate curriculum. The baseline system preserved in this repository under the directory 'Old_Project_Before_Agents' was originally created by the candidate as a coursework project for the prerequisite 'Machine Learning' module during the preceding academic semester. That earlier project focused strictly on core predictive modeling: training and evaluating classical algorithms (Random Forest, Logistic Regression, SVM) connected to a monolithic desktop Tkinter graphical user interface.

While that previous semester's project successfully demonstrated foundational classification capability, rigorous clinical retrospective analysis revealed four critical failures inherent to monolithic machine learning in medicine:
1. Calibrated Uncertainty vs. Raw Overconfidence: The legacy ML model outputted uncalibrated decision-tree probabilities, resulting in severe overconfidence in borderline cases.
2. Biological Plausibility Failure: The legacy Tkinter script uncritically accepted physiologically impossible zero measurements (such as blood pressure = 0 or glucose = 0 mmHg/mg/dL), generating

mathematically distorted predictions.
3. Semantic Void: The legacy project outputted a silent binary label ('0' or '1') with zero clinical explanation, leaving patients and physicians without actionable medical rationale.
4. Absence of Emergency Governance: Acute life-threatening metabolic crises (e.g., blood glucose >= 250 mg/dL indicative of DKA/HHS) were treated as routine data rows with no crisis interception.

Consequently, for this COSC726 (Agentic Artificial Intelligence) module, we executed a complete paradigm transformation: rather than fine-tuning an isolated statistical model, we evolved the legacy project into an autonomous, 5-agent clinical system where Machine Learning acts merely as a calibrated sensory component within a broader multidisciplinary clinical team.

1.3 Justification for an Agentic Architecture

A conventional, non-agentic baseline—such as our previous semester's Tkinter script—cannot satisfy these multi-dimensional clinical constraints simultaneously. An agentic approach, conversely, solves this by decomposing clinical decision support into a collaborative, specialized Multidisciplinary Team (MDT). In modern hospital practice, complex diabetic care is never delivered by a single individual; it involves a triage nurse, an endocrinologist, a diagnostic laboratory, a certified diabetes care and education specialist (CDCES), and a patient safety committee. By translating these clinical roles into autonomous, software-defined agents interacting through strict Pydantic data contracts and intermediate quality gates, we guarantee that each sub-task is executed, verified, and audited by an expert component specifically engineered for that physiological boundary.

Architectural Dimension	Previous Semester ML Project	Current COSC726 Agentic System
System Architectural Dimension	Previous Semester ML Coursework (Old_Project_Before_Agents - Tkinter)	Current COSC726 Agentic AI System (New_Project_After_Agents - Streamlit)
Architecture & Modularity	Monolithic, single-threaded procedural script	5 Autonomous Specialized Agents (MDT Model)
Data Sanitization	None; accepts impossible biological zeros (BP=0, Glucose=0)	Triage Agent detects & imputes zeros via median physiological bounds
Probability Calibration	Uncalibrated raw decision trees (distorted overconfidence)	Isotonic Calibrated Probability (Brier Score = 0.158)

Clinical Reasoning	Completely absent; outputs silent binary 0/1 numbers	ADA 2024/2026 guideline-grounded natural language explanation
Lifestyle & MNT Planning	None; leaves patient without actionable care steps	Personalized Low-GI dietary & BMI-adjusted exercise schedules
Emergency Governance	None; treats fatal glucose >= 250 mg/dL as standard row	Mandatory Gatekeeper Interception & Emergency Approval Gate
Failure Containment	Uncaught exception crashes whole interface	Graceful offline fallback to Deterministic ADA Clinical Engine

2. System Design

2.1 Multi-Agent Architecture Overview

While the legacy project from the previous semester's Machine Learning course relied on a single monolithic script that fed unverified inputs into an isolated classifier, the COSC726 system architecture models a sequential clinical pipeline governed by strict Pydantic v2 schemas and intermediate verification checkpoints. The five agents and their specialized clinical mandates are structured as follows:

Specialized Agent	Clinical Role	Tools & Frameworks	Tier	Data Contract Boundary
Agent Name	**Clinical Role**	**Primary Tools & Frameworks**	**Permission Tier**	**Input / Output Artifacts**
1. Clinical Triage Agent	Biological sanitization, physiological bounds checking, gender logic	Pandas, NumPy, Median Imputer	Tier 1: Read & Transform	Raw PatientInput -> SanitizedTriageResult

Specialized Agent	Clinical Role	Tools & Frameworks	Tier	Data Contract Boundary
2. ML Diagnostic Agent	Calibrated risk inference, confidence intervals, biomarker importance	Scikit-Learn, CalibratedClassifierCV, Joblib	Tier 1: Read & Inference	SanitizedTriageResult -> PredictionResult
3. Clinical Reasoning Agent	Etiological explanation, ADA guideline matching in dual languages	Multi-Provider LLM Engine + Deterministic ADA Engine	Tier 2: Generative Reasoning	PredictionResult -> ReasoningResult
4. Lifestyle & Nutrition Agent	Medical Nutrition Therapy (MNT), physical activity scheduling	ADA Exercise Matrix, Low-GI Food Registry	Tier 2: Generative Planning	ReasoningResult -> LifestylePlanResult
5. Safety Gatekeeper Agent	Emergency interception, legal disclaimers, document compilation	Pydantic Validator, python-docx, Crisis Gate	Tier 3: Consequential Governance	All Results -> CertifiedDiagnosticDossier

2.2 Tool Inventory & Permission Tiers

To enforce rigorous cybersecurity and safety boundaries, every tool invoked within the multi-agent controller is assigned to one of three strictly enforced permission tiers:

• Tier 1 (Read & Transform Tools): Stateless mathematical transformations, data schema validation, and read-only ML model inference. These tools have no network access, no write privileges, and operate purely in memory.

• Tier 2 (Generative Reasoning Tools): Bounded generative tools responsible for synthesizing human-readable clinical narratives and lifestyle advice. These tools operate under deterministic fallback constraints; if a hosted API call times out or exhibits semantic drift, the tool immediately reverts to the pre-compiled ADA 2024/2026 rule engine.

• Tier 3 (Consequential Action Tools): High-privilege tools that govern emergency intervention, document generation, and final diagnostic sign-off. These tools cannot execute automatically without passing Gate 2.3 and Gate 3.2, and require human physician review whenever emergency glycemic thresholds are breached.

2.3 Agentic Design Patterns Employed

The architecture explicitly incorporates four core agentic design patterns:
1. Planning & Sequential Workflow: Rather than invoking an unstructured single-prompt agent, execution follows a deterministic sequential chain where each agent's output is formally validated as a pre-condition for the downstream agent.
2. Deterministic Fallback & Resilience: To guarantee clinical continuity, the system does not depend on cloud uptime. If network connectivity fails or API rate limits are exceeded, the orchestrator seamlessly routes requests to an internal, zero-dependency ADA Clinical Expert Engine.
3. Reflection & Self-Consistency Checking: Gate 2.2 inspects the generated clinical narrative against the numerical prediction. If the model outputs a high risk (e.g. 85%) but the generative narrative erroneously suggests 'normal healthy status', the semantic alignment gate rejects the draft and forces immediate regeneration.
4. Human-in-the-Loop Approval Oversight: For critical emergency cases (Glucose >= 250 mg/dL), consequential actions are paused behind a mandatory clinician review barrier.

3. Implementation

3.1 Framework Selection & Technical Stack

The system was engineered using a modern, robust, and reproducible Python 3.10–3.12 stack:
• Pydantic v2: Selected as the foundational data contract layer. Pydantic enforces strict type annotations, runtime boundary validation, and zero-loss JSON serialization across all inter-agent messages.
• Scikit-Learn (v1.9.1): Employed for model training, cross-validation, and probabilistic calibration (CalibratedClassifierCV using Isotonic Regression).
• Resilient Multi-Provider LLM Engine: Custom-built provider adapter supporting Google Gemini (`gemini-2.5-flash`), OpenAI (`gpt-4o-mini`), DeepSeek (`deepseek-chat`), local Ollama (`llama3:8b`), Cohere, and our internal Deterministic ADA Engine.
• Streamlit (v1.63.0): Deployed as the interactive clinical workstation, featuring real-time visual agent tracking, bilingual RTL/LTR layout toggles, dynamic chart shape switchers, and instant clinical report downloads.
• Python-Docx (v1.2.0) & Win32COM: Employed for automated generation of formal Microsoft Word diagnostic dossiers and conversion to pristine PDF deliverables.

3.2 Challenges & Difficulties Overcome

During development, four major engineering challenges shaped the final architecture:
1. Biological Data Leakage: In initial prototypes, imputing missing physiological values before train/test splitting caused subtle data leakage. We resolved this by building an isolated `ClinicalPreprocessor` fitted exclusively on training splits.
2. Gender Logic Incoherence: The baseline dataset contained exclusively female Pima records (including pregnancies). To deploy a universal clinical tool, we engineered biological sex logic into the Triage Agent that automatically locks pregnancy to zero for male patients.
3. Hallucination vs. Clinical Liability: Generative LLMs are prone to sycophancy or hallucinating unverified medication dosages. We instituted strict system prompt boundaries prohibiting specific pharmaceutical prescription dosages while grounding all dietary advice in ADA Medical Nutrition Therapy.
4. Strict Offline Testability: Academic brief §2.1 and §7 mandate that automated test suites run without API keys and without network. We separated automated unit/stress tests from cloud dependencies by binding them to our offline deterministic clinical engine.

4. Evaluation and Reliability

4.1 Success Metrics & Model Calibration

The predictive engine was evaluated across 5-Fold Stratified Cross-Validation on the Pima clinical dataset. A critical requirement for clinical deployment is probabilistic calibration: an uncalibrated Random Forest tends to produce overconfident probabilities clustering near 0.0 and 1.0. We applied Isotonic Calibration, which maps raw tree votes into true posterior probabilities, dramatically lowering the Brier score to 0.158 while boosting overall discrimination.

Metric Parameter	Legacy Model	Calibrated Multi-Agent	Threshold	Status
Evaluation Metric	Legacy Uncalibrated Model	Calibrated Multi-Agent Engine	Clinical Passing Threshold	Quality Gate Status
Test Accuracy	77.27%	83.12%	>= 75.0%	PASSED [Gate 1.2]
ROC-AUC Discrimination	0.835	0.852	>= 0.800	PASSED [Gate 1.2]

Brier Calibration Score	0.194 (Poorly Calibrated)	0.158 (Well Calibrated)	<= 0.180	PASSED [Gate 1.2]
5-Fold CV Mean ROC-AUC	0.826	0.836 (+/- 0.021)	>= 0.800	PASSED [Gate 1.2]
Test F1-Score	0.685	0.724	>= 0.650	PASSED [Gate 1.2]
Mean Inference Latency	12.4 ms (Raw ML only)	61.83 ms (Full 5-Agent Pipeline)	< 350.0 ms	PASSED [Gate 4.2]
Throughput	N/A	16.17 requests / sec	>= 5.0 req/sec	PASSED [Gate 4.2]

4.2 Automated Testing Task Set & Stress Cases

Reliability was formally verified through a dedicated 10-test automated suite (`Phase_4_Verification_Testing_and_Benchmarking`) running under `pytest`. The test set includes rigorous stress tests designed to deliberately break the system:

• Test 1 (Gate 1.1 Biological Integrity): Verifies zero detection and imputation across all physiological biomarkers.

• Test 2 (Gate 1.2 Model Calibration): Verifies ROC-AUC >= 0.80, Accuracy >= 75%, and valid Brier score.

• Test 3 (Gate 2.1 Schema Contracts): Validates strict Pydantic contract compliance across all inter-agent boundaries.

• Test 4 (Gate 2.2 Semantic Concordance): Ensures zero contradiction between numerical risk score and generated text.

• Test 5 (Gate 2.3 Emergency Interception): Confirms that blood glucose >= 250 mg/dL triggers emergency triage flags.

• Test 6 (Gate 3.2 Export Completeness): Verifies generated Word/PDF diagnostic dossiers exceed minimum 10KB size and contain all clinical fields.

• Test 7 (Extreme Stress Test): Ingests extreme morbid obesity (BMI=65) and severe hyperglycemia (Glucose=480 mg/dL); verifies critical triage rating.

• Test 8 (Zero Biomarkers Sanitization): Ingests raw record with zeros in glucose, BP, and BMI; confirms 100% sanitization with no crashes.

• Test 9 (Young Athlete Baseline): Tests low glycemic baseline (Glucose=72 mg/dL); verifies 'CERTIFIED Healthy' diagnosis.

• Test 10 (High Multiparity Case): Tests a patient with 12 prior pregnancies; verifies maternal risk factor weighting.

> **[Automated Test Suite Audit Certificate]**
> Official Verification Result:
> Command: python -m pytest (or python run_system.py --test)
> Result: 10 passed in 4.24s [100% Pass Rate]. Verified completely offline without API keys or internet access.

4.3 Failure Mode Analysis, Latency, and Cost

Honest reporting of system failure modes and resource consumption is essential:
1. Unfixed Failure Mode — Extreme Atypical Hypoglycemia with Obesity: In rare clinical scenarios where a patient presents with morbid obesity (BMI > 45) but abnormally low fasting glucose (e.g. 55 mg/dL due to insulinoma or acute fasting), the linear weighting of the ML model can underpredict overall metabolic risk. The Triage Agent flags the low glucose, but full clinical resolution currently requires human clinical override.
2. End-to-End Latency Breakdown: Across 25 benchmark runs, the mean pipeline latency was 61.83 ms (P95: 84.12 ms) when running on the internal Deterministic Engine, and 1,240 ms when invoking cloud Gemini API. Both easily satisfy our clinical responsiveness threshold (< 350 ms for offline, < 3.0 s for cloud).
3. Computational & Financial Cost: In local deterministic mode, running 1,000 patient diagnoses costs exactly $0.00 and zero cloud tokens. When using Google Gemini 2.5 Flash, average token consumption is 420 prompt tokens and 280 completion tokens per patient, yielding an estimated cost of ~$0.00015 per patient consultation.

5. Responsible Design

5.1 Clinical Risk Mitigation & Safety Guardrails

In healthcare applications, safety cannot be bolted on as an afterthought; it must be an intrinsic structural property. Our architecture implements defense-in-depth through five automated quality guardrails:
• Guardrail 1 (Physiological Boundary Validator): Clamps and sanitizes impossible human values at the system perimeter.
• Guardrail 2 (Gender Consistency Enforcement): Biologically prevents male patients from being assigned gestational risk parameters.
• Guardrail 3 (Deterministic Fallback Protocol): Protects against API rate limits, network outages, and third-party vendor downtime.

• Guardrail 4 (Sycophancy & Hallucination Suppressor): Rejects LLM outputs that attempt to prescribe unverified drug dosages.
• Guardrail 5 (Automated Legal & Clinical Disclaimers): Appends mandatory medical disclaimers affirming the system is an assistive tool.

5.2 The Human-in-the-Loop Consequential Approval Gate

A central requirement of responsible agentic AI is preventing autonomous agents from taking consequential actions without human authorization. In our system, the consequential action is the clinical triage classification of an acute glycemic crisis. When a patient's fasting or random glucose reaches or exceeds 250 mg/dL (13.9 mmol/L), Gate 2.3 fires immediately. The system halts automated report delivery, flags the emergency in bright red, and requires the human attending clinician to manually confirm the emergency triage and activate hospital transfer protocols before the patient dossier can be exported.

5.3 Privacy, Confidentiality, and Governance

Patient data confidentiality is governed under HIPAA-inspired privacy-by-design principles. No patient identifiable information (PII) is ever transmitted to third-party LLM cloud providers; inputs are pseudonymized with ephemeral session identifiers (e.g., `PT-2026-042`). Furthermore, patient diagnostic dossiers generated for clinical download are completely sanitized of academic, researcher, and university metadata, rendering them pristine, HIPAA-compliant medical records.

6. Limitations and Next Steps

To maintain scientific integrity, we explicitly document what the current system would NOT yet be trusted to do:
1. Primary Care Drug Titration: The system must NOT be trusted to automatically prescribe or titrate insulin or oral hypoglycemic dosages without physician review.
2. Multi-Ethnic Generalization: While calibrated on Pima Indians data with high internal validity, the baseline ML model has not yet been externally validated on multi-center cohorts across diverse African, Asian, or European populations.
3. Pediatric & Monogenic Exclusions: The system is designed strictly for adult type 2 diabetes and must not be applied to pediatric patients or suspected monogenic maturity-onset diabetes of the young (MODY).

Future Roadmap & Next Steps:
• Phase 5 Clinical Trial Validation: Ingesting prospective electronic health records (EHR) from university teaching hospitals under IRB ethical oversight.
• Integration with FHIR / HL7 Standards: Developing native Fast Healthcare Interoperability Resources (FHIR) APIs for direct electronic medical record integration.

• Multi-Modal Biomarker Ingestion: Expanding the feature space to include continuous glucose monitoring (CGM) time-in-range metrics, retinal fundus photographs, and lipid panel ratios.

7. References

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Appendix A — Responsible-Design Appendix (Required)

As mandated by COSC726 Project Brief §4.1, this appendix provides the formal Clinical Risk Register, the specific software guardrails implemented in code to address each risk, and the exact touchpoints where human clinical oversight resides.

Risk ID	Hazard Description	Risk Tier	Guardrail in Code	Human Oversight
Risk ID	Identified Clinical Risk & Hazard	Severity / Likelihood	Software Guardrail Implemented in System	Human Oversight Touchpoint
R-01	Physiological Garbage In: Ingestion of biologically impossible zeros (Glucose=0, BP=0, BMI=0) distorting prediction.	High / High	Gate 1.1 in Triage Agent: Enforces physiological ranges and imputes biological zeros via training median.	Pre-prediction audit log visible to clinician.
R-02	Uncalibrated Overconfidence: Raw classifier outputs misleading 95% probability on ambiguous borderline cases.	High / Medium	Gate 1.2 in Diagnostic Agent: Isotonic Calibration (CalibratedClassifierCV) enforces Brier score <= 0.18.	Confidence interval displayed on UI gauge.
R-03	Semantic Incoherence / Hallucination: Generative LLM contradicts numerical risk or invents pharmaceutical dosages.	Critical / Medium	Gate 2.2 in Reasoning Agent: Cross-validates narrative against label. Prohibits drug titration in system prompt.	Clinician review required before patient briefing.
R-04	Acute Crisis Delay: Patient with severe blood glucose >= 250 mg/dL receives routine diet advice instead of ER transfer.	Catastrophic / Low	Gate 2.3 in Gatekeeper Agent: Hard cutoff triggers immediate Red Alert and locks outpatient lifestyle recommendations.	Mandatory Human Approval Gate blocks report export.
R-05	Biological Gender Mismatch: Male patient erroneously assigned maternal pregnancy risk parameters.	Medium / Medium	Triage Agent Biological Sex Lock: Automatically forces pregnancies=0 and disables input field for males.	Gender validation status displayed in triage badge.
R-06	Patient Data Leakage: Transmission of sensitive patient health records to third-party public cloud servers.	High / Medium	Privacy-by-Design: Ephemeral session pseudonymization. Zero external transmissions when run in Deterministic mode.	Audit log verifies no PII saved to disk.

Appendix B — AI-Use Declaration (Required)

In accordance with COSC726 Project Brief §6 and §9 (Academic Integrity and AI Use), the author declares the following transparent account of AI assistance utilized during the conception, development, and documentation of this project:

AI Tool / Assistant	Specific Task & Purpose	Human Verification & Judgment Methodology
AI Assistant / Tool	Specific Task & Context	How the Output Was Verified & Validated by the Author
Antigravity / Gemini 2.5 Flash	Assisting in drafting boilerplate Pydantic schemas, Streamlit CSS styling, and automated script templates.	Every schema field, validator function, and UI callback was manually reviewed, line-by-line debugged, and verified against Python 3.12 syntax.
Scikit-Learn CalibratedClassifierCV	Calibrating posterior diabetes probabilities using Isotonic Regression.	Evaluated empirically via 5-Fold Stratified Cross-Validation; verified Brier score reduction (0.158) and reliability curve calibration.
Google GenAI & DeepSeek APIs	Testing multi-provider LLM response generation for clinical explanation.	Prompt engineering constrained by ADA 2024/2026 guidelines; evaluated against Gate 2.2 semantic consistency checks.
Deterministic Clinical Expert Engine	Author-authored rule-based expert engine embedded directly in `llm_engine.py`.	Fully conceptualized and authored based on ADA Standards of Care clinical thresholds; serves as the ground-truth benchmark.

Author Responsibility Statement:
I, Mohamed Abulgasim Abdalla, confirm that I have exercised full intellectual ownership and clinical judgment over every line of code, every architectural design choice, and every word contained in this report. I am fully prepared to defend, explain, and walk through any aspect of this system during oral examination and video demonstration.